

How can architecture survive in a world where the ground itself can no longer support life?

"The sky has not been blue in years. Where cities once stood, only their bones remain — smokestacks still exhaling into a sky that forgot how to clear, rivers turned to cracked memory, forests reduced to charcoal silhouettes reaching for a sun they can no longer feel. This is what industry left behind when it finished taking everything it wanted."

EARTH // YEAR 2157

Industrial expansion without ecological balance transformed the Earth into a hostile landscape. Toxic air, contaminated soil, and collapsing ecosystems rendered cities uninhabitable and natural resources nearly extinct. In this new reality, survival depends not on rebuilding the past, but on adapting to a world in constant decline. Cities became ruins, ecosystems vanished, and permanent settlements were abandoned as the land beneath them turned uninhabitable.

In this world, survival no longer depends on where you live—it depends on your ability to keep moving.

THE VANCE FAMILY // THE MISSION

Mr. and Mrs. Vance were once ecological scientists, custodians of Earth's last genetic archives and native seed banks. When governments abandoned restoration for survival, they refused to let that knowledge die — transforming their home into a mobile laboratory, carrying their children and the planet's last chance of renewal across the wreckage it left behind.

THE INDIAN TIMES (Imaginary)

GREEN GRAVES: INDIA'S LAST FORESTS OFFICIALLY DECLARED ECOLOGICALLY DEAD.



ND | Environment Bureau | July 18, 2153

India has officially lost its final self-sustaining natural forest, according to the National Ecological Survey released yesterday. Once-thriving ecosystems across the Western Ghats, Himalayas and central forests have now been reclassified as "biologically inactive landscapes" after decades of industrial pollution, groundwater depletion and rising atmospheric toxicity.

Researchers report that more than 96% of native tree species have failed to regenerate naturally over the past twenty years. Soil samples reveal heavy-metal contamination extending several meters below the surface, preventing seed germination even after repeated restoration attempts. Once covering nearly one-third of the nation's land, forests have steadily disappeared under decades of industrial expansion, mining, urban sprawl, and rising atmospheric toxicity. Soil samples collected across former biodiversity hotspots reveal heavy-metal contamination, acidification, and microbial collapse, preventing even preserved native seeds from germinating naturally.

"Trees no longer die because they are cut down," explained Dr. Anika Rao of the National Ecology Council. "They die because the Earth beneath them has forgotten how to support life." Authorities have transferred the remaining viable seed collections to secure biological vaults, while restoration efforts increasingly depend on mobile ecological expeditions capable of carrying and reintroducing life into damaged landscapes.



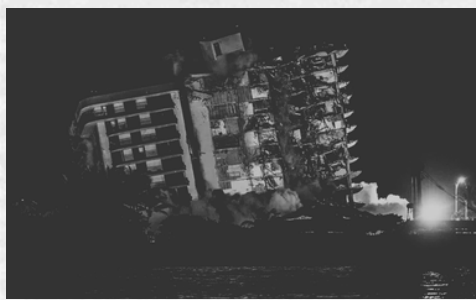
EARTH NEWS NETWORK (Imaginary)

PERMANENT HOMES ABANDONED AS TOXIC LANDSCAPES FORCE A NOMADIC FUTURE

Global Climate Desk | July 21, 2154

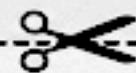
The Ministry of Civil Resilience has confirmed that permanent settlements across multiple regions have been declared uninhabitable following continued soil toxicity, collapsing infrastructure and unpredictable environmental conditions.

Entire cities now lie abandoned beneath layers of industrial ash, while toxic dust storms regularly force surviving communities to relocate. Traditional housing has become increasingly obsolete as stable ground becomes scarce and access to clean water continues to decline. Across continents, abandoned skylines stand as silent reminders of an industrial civilization that exhausted the ecosystems upon which it depended.



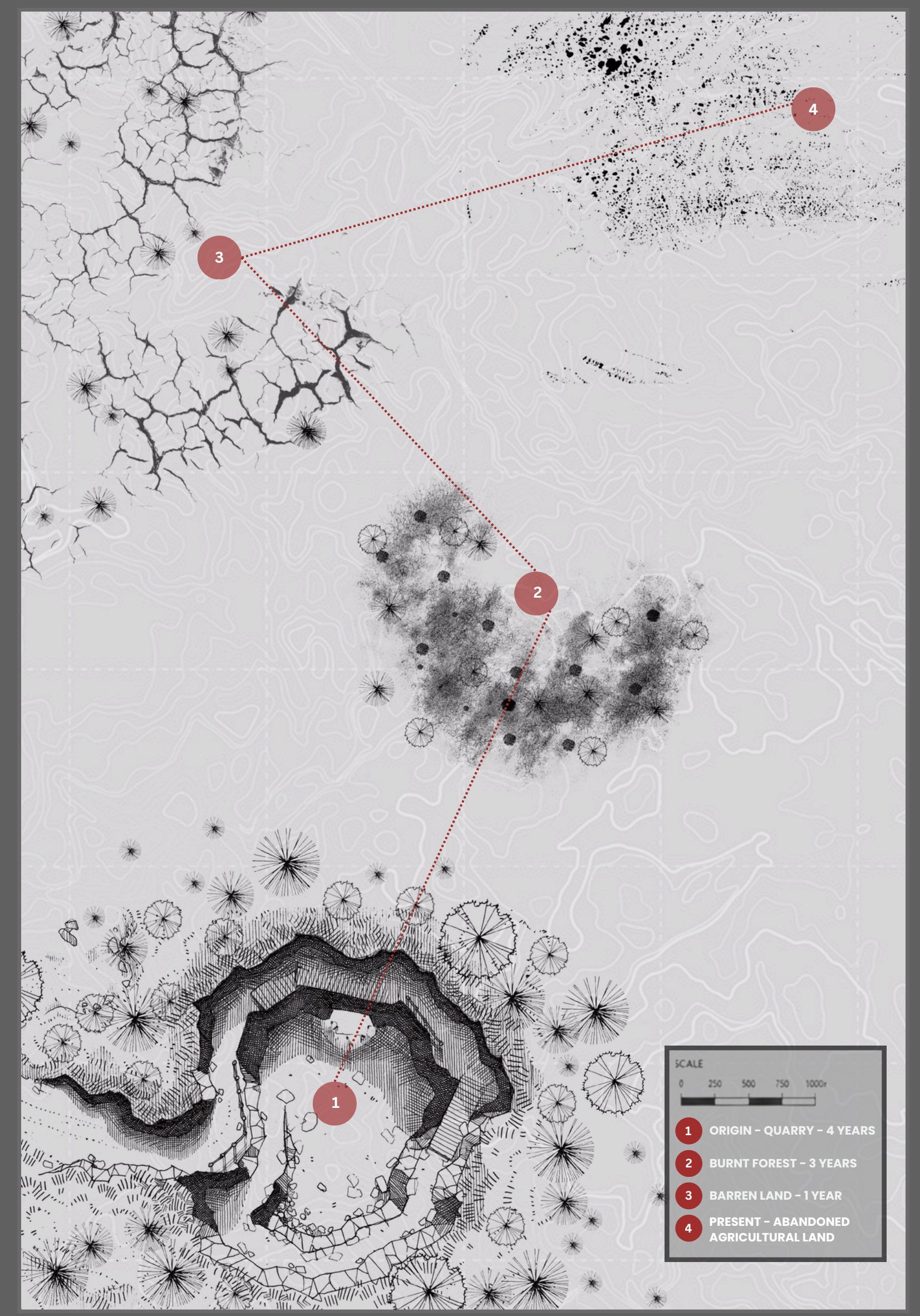
"Home is no longer defined by place," one researcher noted. "It is defined by the ability to survive."

Communities have been forced to adopt mobile, self-sustaining habitats that combine shelter, food production, water purification, and renewable energy into autonomous living systems capable of traversing unstable terrain. Engineers and environmental scientists are now developing mobile self-sustaining habitats capable of carrying food production, water purification and ecological restoration systems. These experimental dwellings represent a fundamental shift in architecture—from buildings fixed to the land to structures designed to move with it.



THE VANCE EXPEDITION // TERRITORY LOG

FIELD SURVEY SURVEY DEPT - THE VANCES ISSUE NO. 07/2157



SITE ANALYSIS // WORLD CONDITIONS

THE COLLAPSE

No single event caused this. What remains is not one wasteland, but many kinds of dying, layered across a continent that industry touched everywhere and healed nowhere. There is no fixed site here — only a world in varying stages of ruin, and a house that moves through all of them.

BIOMES ENCOUNTERED

- Abandoned industrial valleys — dead factory zones, machines stopped, nothing left running
- Dried lake beds — former lakes, now cracked salt flats, no water
- Toxic wetlands — poisoned marsh/swamp, stagnant, nothing survives in it
- Desertified plains — once-fertile land turned to dry dust
- Burnt forests — repeatedly fire-damaged woodland, no time to regrow
- Former agricultural land — old farmland, soil exhausted, abandoned fields
- Collapsing mining regions — old quarries/mines, terraced pits, oldest damage

CLIMATE & ATMOSPHERE

Conditions shift biome to biome — ash haze over burnt land, mineral dust across the plains, chemical taint near old industry. Light is rarely direct, filtered through haze. Rainfall is inconsistent; near old mines, it falls acidic.

WATER & GROUND

Water is present but rarely clean — stagnant near collapsed industry, absent across dried lake beds and desert plains. Ground stability varies: cracked plains, ash-laden forest floors, terraced quarry edges.

This map shows one segment of a longer expedition — each stop a different wound, met with a different answer.

- Stop 1** — The Quarry. Where our journey began. Four years gone, and the forest has already reclaimed what the mining stripped bare. This is the proof that recovery is possible — not fast, but certain.
- Stop 2** — The Burn. Scorched ground, three years past. New growth is fragile here still, but it holds.
- Stop 3** — The Flats. Dry, cracked, barely a year since we passed. Too soon to call it healed — only beginning.
- Stop 4** — Present Position. Abandoned agricultural land beneath us now. This is where the work starts again. Distance traveled: unrecorded. Time is measured differently out here — not in kilometers, but in what's left growing behind us.



"The world no longer has places worth settling, so the Vance family has chosen a different purpose. Their home never stays still because every destination is another wounded landscape waiting to heal. Inspired by the bacteriophage—a microscopic organism that transfers genetic information—the house carries the genetic memory of Earth's forests. Each step injects life into poisoned ground, leaving behind a trail of recovering ecosystems. Home is no longer defined by where the family lives, but by the landscapes they restore."

CONCEPT // PHAGE

THE PRECEDENT

Before the house had a shape, it had a reference: the bacteriophage — a virus that survives by injecting genetic material into a host, rather than consuming it. Studied here at analytical scale, gridded and dissected like a specimen, its anatomy became the starting logic for everything that followed.

WHY THIS ORGANISM ?

Most machines take. The phage inserts — it delivers something, rather than extracting it. That single distinction became the seed of the entire project: a house that doesn't just survive a dying world, but actively re-writes it, one injection, one seed, one footstep at a time.

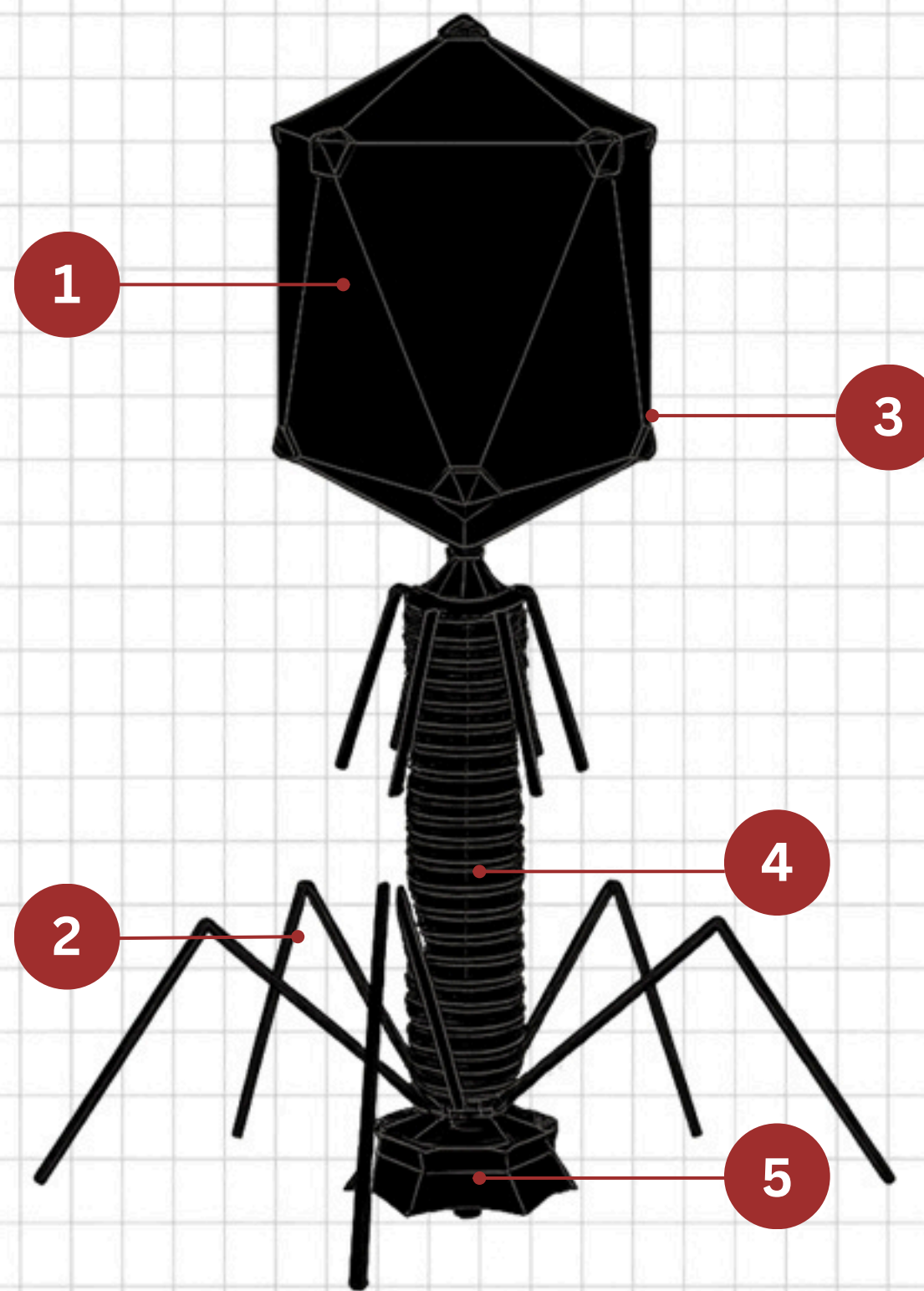
FROM SPECIMEN TO STRUCTURE

The icosahedral capsid becomes the main habitable volume — geometric, faceted, rigid. The tail sheath becomes vertical circulation. The tail fibers become legs. What was microscopic becomes architectural; what injected genetic code now injects seed capsules into dead ground.

THE RESULT

This is not a house shaped like a virus for the sake of spectacle. It is a house that adopts the bacteriophage's logic. Where the phage injects genetic material to transform a bacterium, this structure injects seeds, nutrients, and microbial life into exhausted soil. It moves through a devastated landscape carrying the tools of regeneration within its own body. What was once a mechanism of infection becomes a mechanism of restoration—a living architecture that revives the world with every step.

"It harvests from the sky, sustains from within, and restores the earth beneath its feet."



ANATOMY TO ARCHITECTURE

1. CAPSID (HEAD)

The capsid is the primary living volume of the house. It is organized vertically, with a greenhouse at the top, bedrooms and study spaces on the upper level, communal spaces—including the kitchen, living, dining, TV room, and drawing room—at the center, and a control room at the base for navigation, drone operation, and environmental monitoring.

2. TAIL FIBERS (WALKING LEGS)

The articulated legs allow the house to move across unstable terrain. Each injector foot releases seed capsules, fungal cultures, and beneficial microorganisms into the ground, initiating ecological restoration with every stop.

3. MEMBRANE

The membrane is the house's environmental life-support system. It harvests solar energy, extracts atmospheric moisture, and filters toxic air and contaminated water. These resources are converted into electricity, clean water, and breathable air, then distributed throughout the house via service ducts housed within the structural spine.

4. COLLAR & TAIL SHEATH

Translates directly into the house's structural spine and service core — the same shaft carrying circulation also routes water, air, and power drawn in through the membrane down into the capsid's interior systems.

5. BASEPLATE

The baseplate is the mechanical foundation of the house. It contains the workshop, seed and fungal storage, repair systems, and the injection mechanism that supplies the legs during restoration.



LEGENDS

1. LIFT
2. GREENHOUSE/BIO LAB
3. MASTER BEDROOM
4. STUDY
5. CLOSET
6. DUCTS
7. BEDROOM 1
8. BEDROOM 2
9. LIBRARY AND STUDY
10. LIVING ROOM
11. KITCHEN
12. TV ROOM
13. DINING ROOM
14. DRAWING ROOM
15. LAUNDRY AND STORAGE ROOM

LEVEL 1 – SOCIAL ZONE

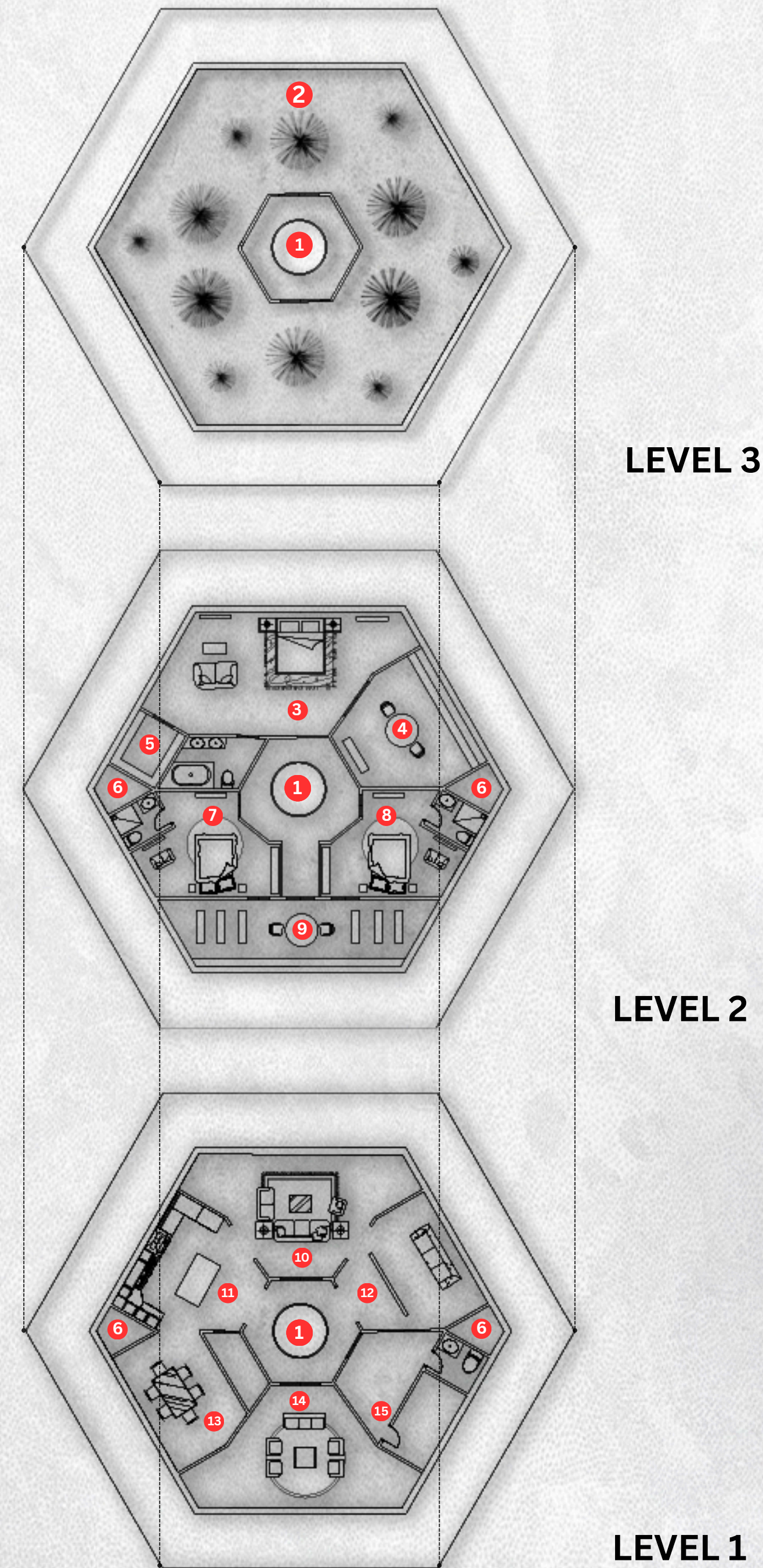
Level 1 serves as the communal area, with the lift at the center connecting all floors. It includes the living room, drawing room, dining room, kitchen, TV room, and a laundry and storage room, while ducts provide essential building services.

LEVEL 2 – PRIVATE ZONE

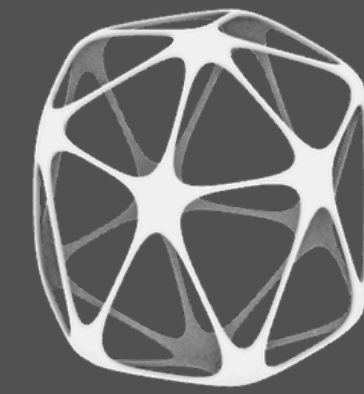
Level 2 is the family's private living area, comprising the master bedroom, two bedrooms, study, library and study, along with closets. The central lift ensures efficient circulation, while ducts distribute utilities throughout the floor.

LEVEL 3 – GREENHOUSE/BIO LAB

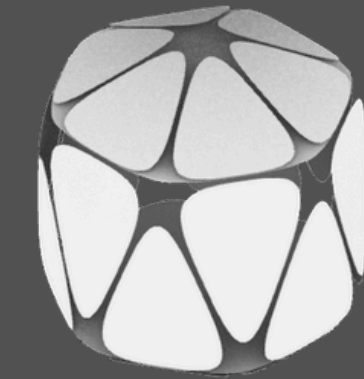
Level 3 is dedicated to the greenhouse and bio lab, accessed through the central lift. This level supports food cultivation, ecological research, and environmental resource harvesting, reinforcing the house as a self-sustaining living system.



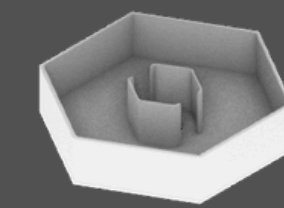
PLANS



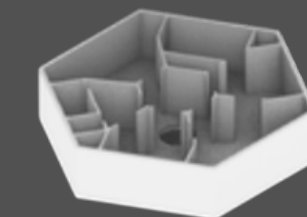
MEMBRANE



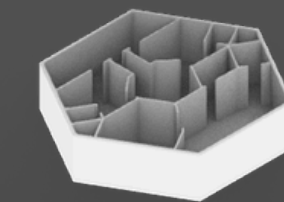
INNER SHELL



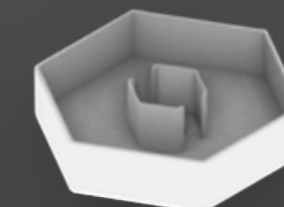
LEVEL 3



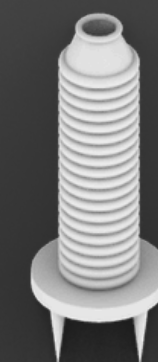
LEVEL 2



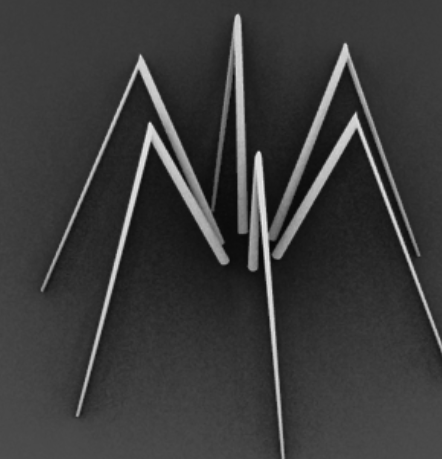
LEVEL 1



LEVEL 0 -
CONTROL ROOM



VERTICAL
SERVICE CORE



WALKING
LEGS

